TEAM 12

MATHS IN ALGORITHMS

LOGIC AND SET THEORY

Algorithms fundamentally rely on logical principles (if-then, and, or, not) and use set theory to group and manage data, which underpins database operations like unions and intersections.

GRAPH THEORY

Graphs are mathematical constructs employed to represent the relationships between various objects.

LINEAR ALGEBRA

Linear algebra involves vectors, matrices, and vector spaces, serving as a fundamental element in modern algorithms, particularly in machine learning and computer graphics.

COMPUTER GRAPHICS

Manipulating 3D objects on a 2D screen uses linear algebra, where operations like rotation, scaling, and translation are performed through matrix multiplications of vertex coordinates.

CALCULUS: THE ENGINE OF OPTIMIZATION

Machine Learning Optimization involves training a model by minimizing a "loss function" that measures prediction inaccuracy, primarily using the Gradient Descent algorithm.

NUMBER THEORY: THE KEY CRYPTOGRAPHY

Public-Key Cryptography: Algorithms such as RSA are based on number theory principles.

PROBABILITY AND STATISTICS: HANDLING UNCERTAINTY

Many algorithms work with uncertain or incomplete data, and probability and statistics help manage this uncertainty.

RANDOMISED ALGORITHMS

Some problems are solved more efficiently with randomness, such as Quicksort, where choosing a random pivot enhances average-case performance.

SPAM FILTERING

Bayesian statistics help spam filters determine the likelihood of an email being spam based on its word content.

MACHINE LEARNING MODELS

Many machine learning algorithms use probabilities; for example, a model may predict a 95% chance of rain tomorrow.

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